Math 445/545. Winter 2026. Practice problems for Midterm 1. Solutions.

January 28, 2026

The problems below illustrate various techniques you are supposed to know for the final. Some of the problems below are more difficult than the ones you will see on the midterm.

Problem 1. Let a be an element of a ring R. Let $S = \{x \in R \mid ax = 0\}$. Show that S is a subring of R.

Solution: We need to check that S with operations inherited from R is a ring. Most importantly we need to check that S is closed under the operations. Let $x, y \in S$. By definition this means $ax = ay = 0$. Hence $a(x + y) = ax + ay = 0 + 0 = 0$, so $x+y \in S$. Also $a(xy) = (ax)y = 0 \cdot y = 0$, so $xy \in S$. Thus S is closed under addition and multiplication.

We need to check that S is abelian group. Clearly, addition in S is associative and commutative (since addition is associative and commutative in a bigger set R). Also S contains zero element since $a \cdot 0 = 0$. Finally if $x \in S$, then $(-x) \in S$ since $a(-x) = -xa = -0 = 0$. Thus S is abelian subgroup of R (with respect to operation of addition).

All the remaining axioms (namely, associativity of multiplication and distributivity) of the ring hold in S since they hold in a bigger set R. Thus S is a subring of R.

Problem 2. Assume that a is a nilpotent element of a unital ring R. Prove that $1 + a$ is a unit of R.

Solution: The element a nilpotent means that $a^n = 0$ for some $n \in \mathbb{Z}_{\geq 1}$. Set $b = 1 - a + a^2 - a^3 + \ldots \pm a^{n-1}$. Then

$$b(1 + a) = b + ba = 1 - a + a^2 - a^3 + \ldots \pm a^{n-1} + a - a^2 + a^3 - a^4 + \ldots \pm a^n = 1.$$

A very similar computation shows that $(1 + a)b = 1$. Thus b is the inverse of $1 + a$ and we are done.

Problem 3. What is characteristic of the ring $\mathbb{Z}_4 \times 4\mathbb{Z}$?

Solution: Let us take element $x = (0, 4) \in \mathbb{Z}_4 \times 4\mathbb{Z}$. Then for any $n \in \mathbb{Z}_{\geq 1}$ we have $nx = (0, 4n) \neq (0, 0)$, that is no multiple of x is zero. Hence the characteristic of the ring $\mathbb{Z}_4 \times 4\mathbb{Z}$ is zero.

Problem 4. What is the number of irreducible polynomials of degree 3 in $\mathbb{Z}_2[x]$?

Solution: The polynomial of degree 3 looks like $x^3 + a_2x^2 + a_1x + a_0$. For an irreducible polynomial $a_0 = 1$ (since otherwise 0 is zero) and $a_2 + a_1 = 1$ (since otherwise 1 is zero). Thus we have 2 possibilities (since non-irreducible polynomial of degree 3 must have a zero).

Problem 5. Compute evaluation $\phi_2(x^{100} + 2x^{31} + 3x^2 + 5)$ when the polynomial is considered over $\mathbb{Z}_{11}$ (so the answer should be some element of $\mathbb{Z}_{11}$).

Solution: By definition of the evaluation homomorphism we have

$$\phi_2(x^{100} + 2x^{31} + 3x^2 + 5) = 2^{100} + 2 \cdot 2^{31} + 3 \cdot 2^2 + 5 \pmod{11}.$$

By Fermat's little theorem we have $2^{10} \equiv 1 \pmod{11}$, so $2^{100} = (2^{10})^{10} \equiv 1^10 = 1 \pmod{11}$ and $2^{30} = (2^{10})^3 \equiv 1^3 = 1 \pmod{11}$ whence $2^{31} = 2 \cdot 2^{30} \equiv 2 \cdot 1 = 2 \pmod{11}$. Hence

$$2^{100} + 2 \cdot 2^{31} + 3 \cdot 2^2 + 5 \equiv 1 + 2 \cdot 2 + 3 \cdot 4 + 5 = 22 \equiv 0 \pmod{11}.$$

Thus 2 is a zero of polynomial $x^{100} + 2x^{31} + 3x^2 + 5 \in \mathbb{Z}_{11}[x]$.

Answer: $\phi_2(x^{100} + 2x^{31} + 3x^2 + 5) = 0$.

Problem 6. In the ring $\mathbb{Z}_{3600}$ find the number of idempotents, nilpotents, units, and zero divisors.

Solution: To find the idempotents we need to solve congruence

$$x^2 \equiv x \pmod{3600}.$$

By the Chinese Remainder Theorem this is equivalent to 3 congruences modulo $2^4, 3^2, 5^2$. Each of these congruences has 2 solutions – 0 and 1 (e.g. $x^2 - x$ is divisible by 25 if either x or $x - 1$ is divisible by 25). Thus we have $2 \cdot 2 \cdot 2 = 8$ solutions of the original congruence.

The nilpotents are precisely classes $[x]$ where x is divisible by 2,3, and 5; equivalently x is divisible by 30. Thus we have $\frac{3600}{30} = 120$ nilpotents.

The number of units is given by the Euler's function $\phi(3600) = \phi(2^4)\phi(3^2)\phi(5^2) = 2^3 \cdot 6 \cdot 20 = 960$.

Finally, the zero divisors are classes $[x]$ where x is not relatively prime with 3600. Equivalently, the zero divisors are all elements which are not units. Thus we have $3600 - 960 = 2640$ zero divisors.

Problem 7. What is the remainder of $x^{100} + 1 \in \mathbb{R}[x]$ when divided by $x^2 - 2x$?

Solution: The remainder $r(x)$ is polynomial of degree < 2 and such that $x^{100}+1 = (x^2 - 2x)q(x) + r(x)$. Thus $r(x) = ax + b$; by plugging in (or applying evaluation homomorphism) $x = 0$ and $x = 2$ we find $b = 1$ and $2a+b = 2^{100}+1$, whence $a = 2^{99}$. Thus $r(x) = 2^{99}x + 1$.

Problem 8. Suppose that R is a ring such that $x^3 = x$ for all $x \in R$. Prove that $6x = 0$ for all $x \in R$.

Solution: Let $x \in R$. Then $x^3 = x$ by the assumptions. Now $y = 2x$ is also in R, so $y^3 = y$. By plugging in $y = 2x$ we get $8x^3 = y^3 = y = 2x$. Now let us combine our equations: $2x = 8x^3 = 8x$. Hence $6x = 8x - 2x = 0$ as required.

Problem 9. (a) Prove that the ring $\mathbb{Z}[\sqrt{2}]$ has infinitely many units.

(b) Prove that the ring $\mathbb{Z}[i]$ has only finitely many units and find them all.

Solution: (a) The element $\sqrt{2}+1$ is a unit since $(\sqrt{2}+1)(\sqrt{2}-1) = 1$. Thus any power $(\sqrt{2}+1)^n, n \in \mathbb{Z}_+$ is a unit; all these powers are distinct as real numbers, so we get infinitely many units.

(b) For any $z \in \mathbb{Z}[i]$ its complex conjugate $\bar{z}$ is also an element of $\mathbb{Z}[i]$. Now assume $z \in \mathbb{Z}[i]$ is a unit, so there is $w \in \mathbb{Z}[i]$ such that $zw = 1$. Taking complex conjugates we get $\bar{z}\bar{w} = 1$. Multiplying two equations we get $z\bar{z}w\bar{w} = zw \cdot \bar{z}\bar{w} = 1$. Now let $z = a + bi, w = c + di$ where $a, b, c, d \in \mathbb{Z}$. Then $zbarz = a^2 + b^2 \in \mathbb{Z}_{\geq 0}$ and $w\bar{w} = c^2 + d^2 \in \mathbb{Z}_{\geq 0}$ and we just proved $(a^2 + b^2)(c^2 + d^2) = 1$. Thus $a^2 + b^2 = 1$ and we have exactly 4 options: $a = \pm 1, b = 0$, or $a = 0, b = \pm 1$. Thus we have precisely 4 units: ± 1 and $\pm i$.

Problem 10. Is the group $\mathbb{Z}_{143}^\times$ cyclic? $(143 = 11 \cdot 13)$.

Solution: By the Chinese remainder theorem we have $\mathbb{Z}_{143}$ is isomorphic to $\mathbb{Z}_{11} \times \mathbb{Z}_{13}$. Hence $\mathbb{Z}_{143}^\times$ is isomorphic to $\mathbb{Z}_{11}^\times \times \mathbb{Z}_{13}^\times$. Since 11 and 13 are primes, we know that the groups $\mathbb{Z}_{11}^{\times}$ and $\mathbb{Z}_{13}^{\times}$ are cyclic of orders 10 and 12. Thus $\mathbb{Z}_{143}^{\times}$ is isomorphic to $\mathbb{Z}_{10} \times \mathbb{Z}_{12}$. In particular, the order of any element of this group is a divisor of $\operatorname{lcm}(10, 12) = 60$. Thus this group has no elements of order 120, so it is not cyclic.